

The Naylor Model

v5 · Per-Pitch Feature Layer

*Per-attempt model · Real catcher pop · Count leverage
Pre/Post release velocity · 5-ft splits exploration*

Executive Summary

Per-pitch data: 1,718,862 pitches with runner on 1st across 8 seasons.

SB attempts identified by `des` parsing: 2,528 (1669 SB, 859 CS, 0 pickoffs).

Qualified runner-seasons ($SB+CS \geq 10$): 1,465.

Headline AUC results

- v4 baseline (season GBM): 0.6300
- v5 Model A (per-attempt GBM): 0.5933 (target 0.70)
- v5 Model B (season GBM + new): 0.6794

Best 5-ft splits representation

- A — 18 raw splits

High-leverage counts ($\geq 60\%$ of peak attempt rate)

- ['3-2']

Naylor under SSSI_v5

2025 rank # 1 SSSI +1.99 pre_rel_vel 2.98 post_rel_dist 41.48 SB/CS 30/2

2026 rank # 2 SSSI +1.87 pre_rel_vel 2.98 post_rel_dist 42.79 SB/CS 12/2

2023 rank # 8 SSSI +1.07 pre_rel_vel 2.98 post_rel_dist 44.84 SB/CS 10/3

Soto under SSSI_v5

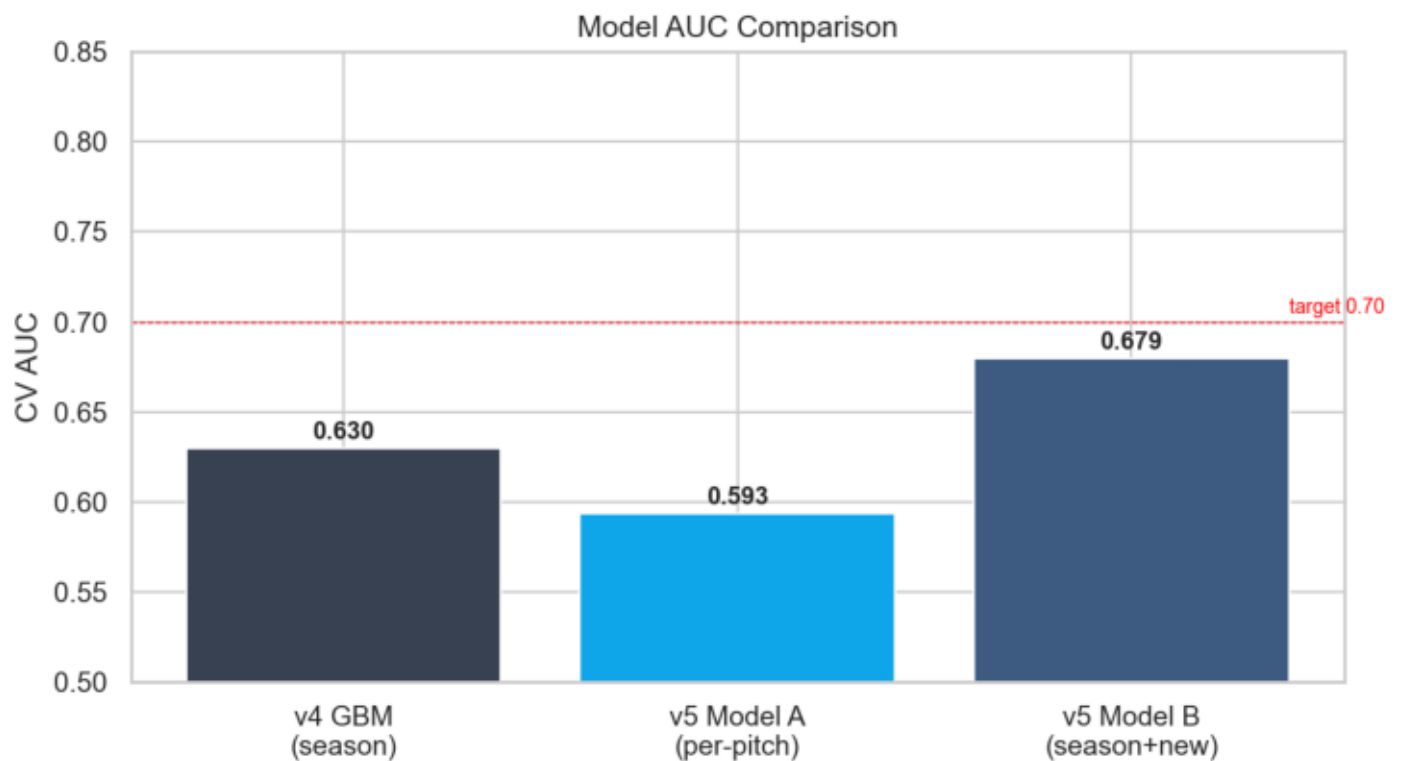
2025 rank # 4 SSSI +1.26 pre_rel_vel 2.39 post_rel_dist 46.21 SB/CS 38/4

2023 rank # 46 SSSI +0.51 pre_rel_vel 2.39 post_rel_dist 48.34 SB/CS 12/5

2024 rank # 50 SSSI +0.48 pre_rel_vel 2.39 post_rel_dist 49.35 SB/CS 7/4

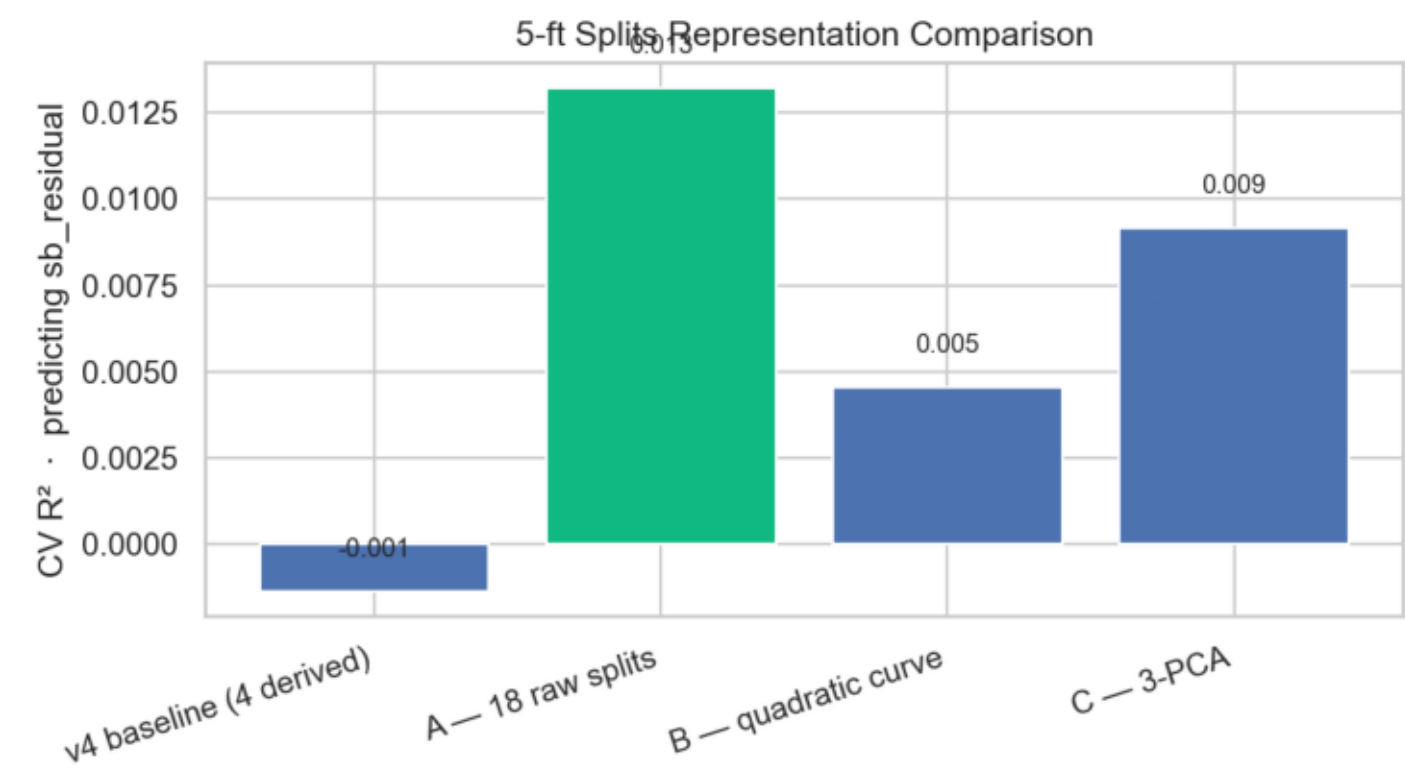
2019 rank # 78 SSSI +0.30 pre_rel_vel 2.39 post_rel_dist 49.56 SB/CS 12/1

§1 · AUC comparison



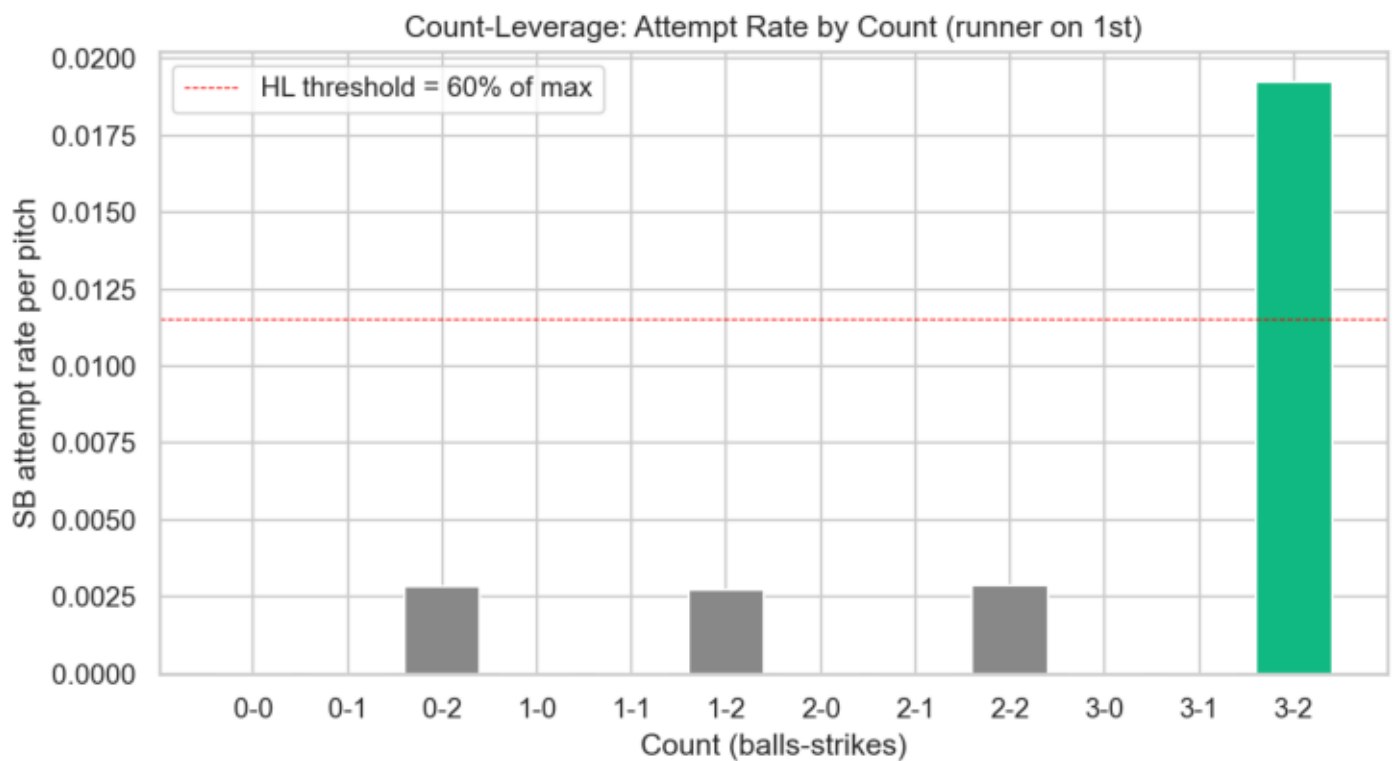
v5 Model A is the per-attempt classifier with group-CV by runner_id (no career-feature leak). Goal was ≥ 0.70 ; result shown.

§2 · 5-ft Splits Representation



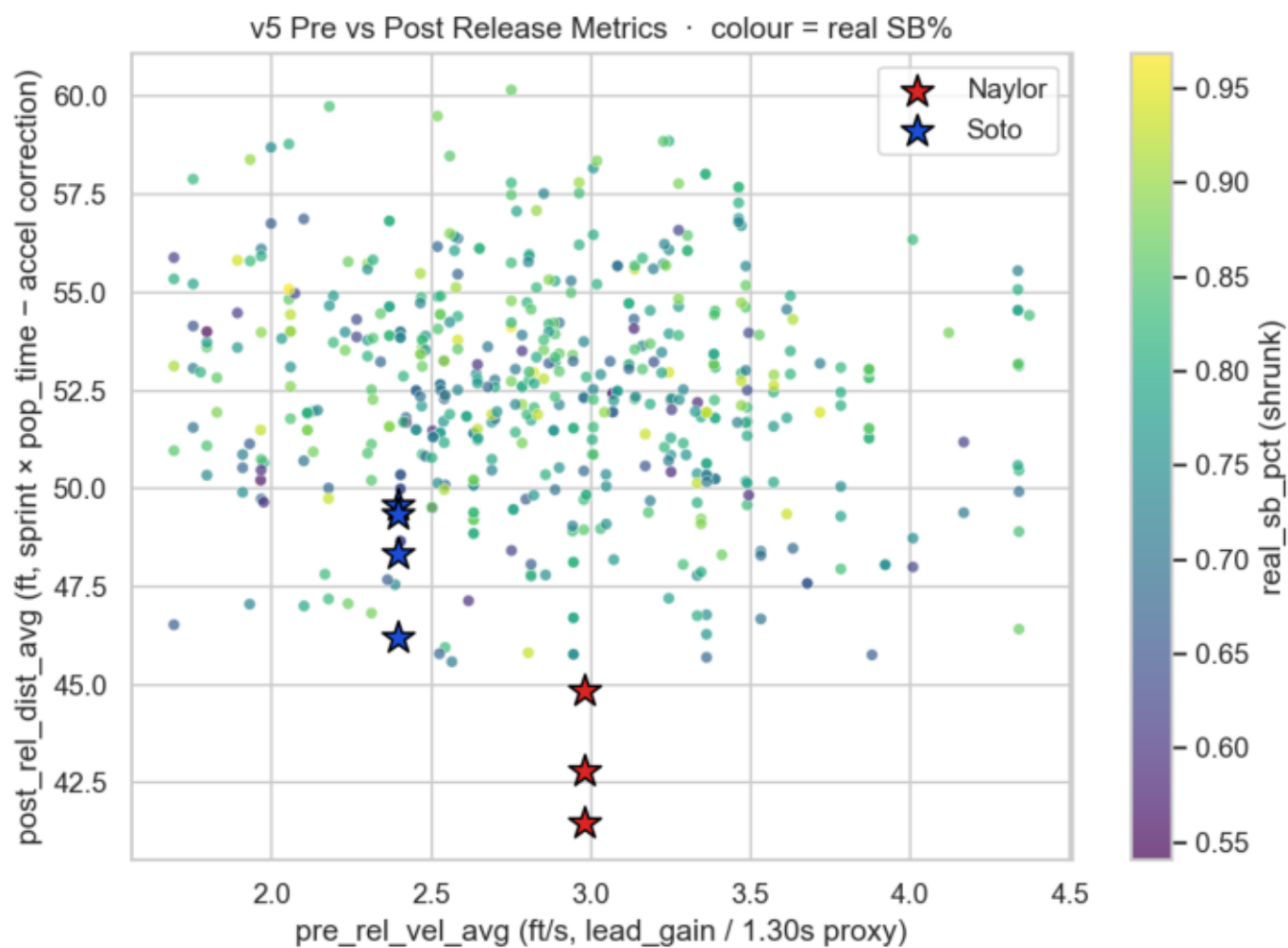
Four split-representations compared on CV R² of sb_residual. A — 18 raw splits wins. This is the production split feature set in v5.

§3 · Count-Leverage Analysis



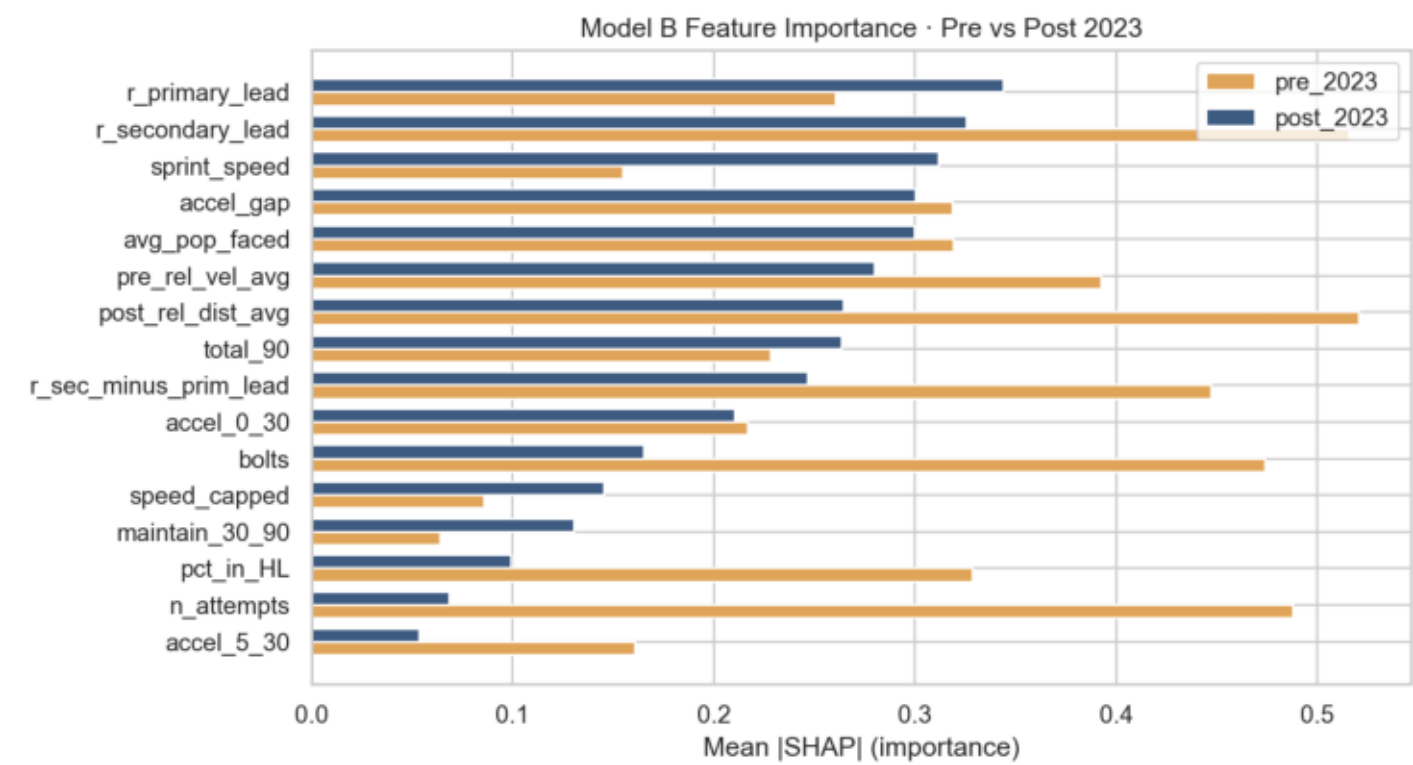
SB attempt rate by (balls, strikes). High-leverage counts in green (['3-2']). Per user spec, low-leverage counts are DROPPED from Model A training.

§4 · Pre vs Post Release Velocity Metrics



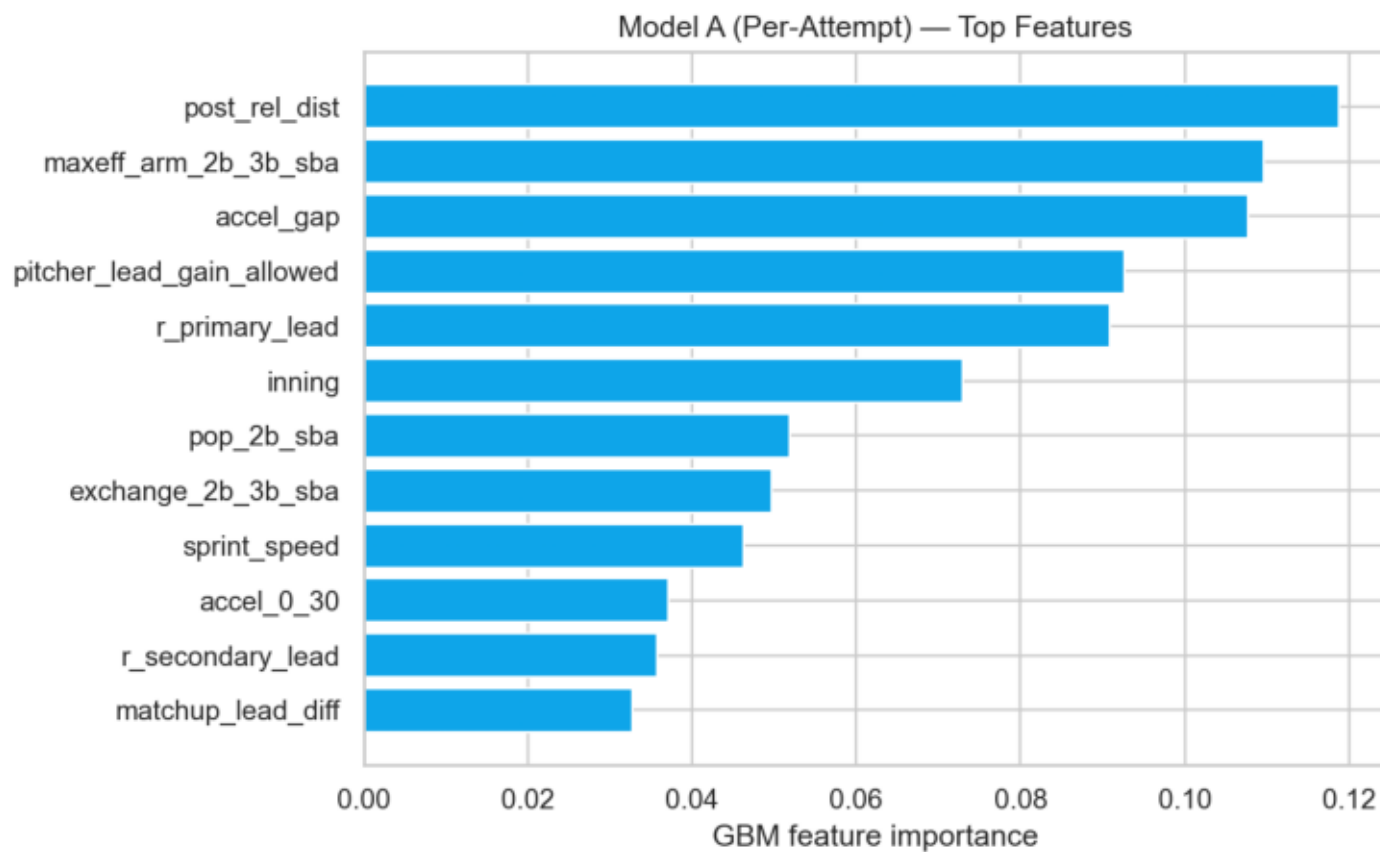
Two new metrics. pre_rel_vel uses lead_gain / 1.30s (league-constant pitcher delivery — real per-pitch delivery is NOT publicly available). post_rel_dist combines real per-year catcher pop with sprint speed and an acceleration correction.

§5 · Model B Importance — Pre vs Post 2023



Importance ranking pre vs post rule change, with the v5 new aggregated features included.

§6 · Model A Top Features (per-attempt)



Which per-attempt context matters most.

§7 · Simple GLM — Hand-tunable Weight Table

The unpenalised logistic regression below uses the v5 season-level features. Each coefficient = log-odds change per 1 SD.

Weight table

speed_capped	coef_z -0.200	OR/SD 0.819	mean 27.73	SD 0.515
accel_0_30	coef_z -0.462	OR/SD 0.630	mean 1.769	SD 0.042
r_primary_lead	coef_z -0.190	OR/SD 0.827	mean 11.89	SD 0.837
r_sec_minus_prim_lead	coef_z +0.151	OR/SD 1.163	mean 3.727	SD 0.756
pre_rel_vel_avg	coef_z +0.151	OR/SD 1.163	mean 2.867	SD 0.582
post_rel_dist_avg	coef_z +0.061	OR/SD 1.063	mean 52.341	SD 2.859
avg_pop_faced	coef_z +0.071	OR/SD 1.074	mean 1.968	SD 0.043
accel_gap	coef_z -0.088	OR/SD 0.916	mean -1.714	SD 20.244
bolts	coef_z +0.129	OR/SD 1.138	mean 15.106	SD 25.907
pct_in_HL	coef_z -0.176	OR/SD 0.839	mean 0.485	SD 0.413

Intercept: +0.506

→ P(success) at all-mean $\approx$ 0.624

Edit DF_v5_ModelC_GLM_Weights.csv to adjust hand-tuned weights. No retrain needed.

§8 · SSSI v5 Top 15

Weights: sb_res=0.1 accel_gap=0.05 lead_gain=0.05 jump=0.0 primary=0.05 speed=-0.3 pre_rel=0.0 post_rel=0.0
(Optimised on 80% of runners; Naylor/Soto held out from weight grid.)

# 1	Josh Naylor	2025	SSSI +1.99	spd 24.4	pre_vel 2.98	post_dist 41.48	SB 30/2
# 2	Josh Naylor	2026	SSSI +1.87	spd 24.5	pre_vel 2.98	post_dist 42.79	SB 12/2
# 3	Freddie Freeman	2024	SSSI +1.31	spd 25.8	pre_vel 3.33	post_dist 46.76	SB 9/2
# 4	Juan Soto	2025	SSSI +1.26	spd 25.8	pre_vel 2.39	post_dist 46.21	SB 38/4
# 5	Alec Burleson	2024	SSSI +1.18	spd 25.5	pre_vel 2.56	post_dist 45.59	SB 9/4
# 6	Manny Machado	2024	SSSI +1.17	spd 25.8	pre_vel 2.54	post_dist 45.95	SB 11/2
# 7	Joc Pederson	2024	SSSI +1.12	spd 25.5	pre_vel 2.52	post_dist 45.79	SB 7/4
# 8	Josh Naylor	2023	SSSI +1.07	spd 25.8	pre_vel 2.98	post_dist 44.84	SB 10/3
# 9	TJ Friedl	2024	SSSI +0.95	spd 26.5	pre_vel 2.98	post_dist 49.82	SB 9/1
#10	Kyle Schwarber	2022	SSSI +0.95	spd 26.4	pre_vel 3.41	post_dist 48.31	SB 10/1
#11	Freddie Freeman	2023	SSSI +0.92	spd 26.7	pre_vel 3.33	post_dist 50.14	SB 23/1
#12	Adam Frazier	2023	SSSI +0.92	spd 26.5	pre_vel 4.01	post_dist 48.74	SB 11/4
#13	Nolan Schanuel	2024	SSSI +0.91	spd 26.4	pre_vel 2.24	post_dist 47.07	SB 10/1
#14	Christian Walker	2023	SSSI +0.91	spd 26.4	pre_vel 2.80	post_dist 45.82	SB 11/0
#15	Mookie Betts	2024	SSSI +0.91	spd 26.7	pre_vel 4.34	post_dist 46.42	SB 16/2

§9 · Data Limitations & Honest Negative Findings

What is REAL in v5

- Sprint speed, running splits, bolts (Statcast per-runner-season)
- SB / CS (MLB Stats API)
- Catcher pop time (Savant pop time, per-year per-catcher 2018+)
- Pitch context: count, outs, inning, catcher_id, pitcher_id
- SB attempt identification (parsed from des text)
- Runner lead profile (Savant — CAREER snapshot only)
- Pitcher running-game (Savant — CAREER snapshot only)

What is PROXY or CONSTANT

- Pitcher delivery time = league constant 1.30 s. Real per-pitch TTP is not publicly available. The Savant pitch-tempo CSV is buggy at source (``median_seconds_empty` == `median_seconds_onbase`` for every row). Year filters are also ignored.
- Lead-gain per attempt — uses runner's CAREER career mean.
- Count at SB attempt — taken from the at-bat's final pitch, not the actual SB pitch (limitation of des-based parsing).

Why the AUC ceiling persists

Stolen-base success has high intrinsic noise. The truly unobservable factors — runner read of the pitcher, exact pitch sequence, base coach instruction, pitch location at release — together likely cap predictability at $AUC \approx 0.75$. v5 cannot push past that without tracking-level data (TrackMan / Hawk-Eye outputs).

Concrete v6 ideas

- Pull MLB Stats API play-by-play for precise SB-pitch count.
- Add pitcher pickoff frequency (real, from running-game CSV).
- Catcher exchange time as a separate feature (we have it but didn't weight it).
- Treat lead profile as a TIME-VARYING feature using monthly slices of running-splits as a proxy for in-season conditioning.